

SOLITON-SHAPED POTENTIALS AND THE GENERALIZED POWER-SERIES METHOD

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ABSTRACT. With the double-well potential appearing in the soliton theory, the one-dimensional Schrödinger equation is solved exactly in terms of the quasihypergeometric series. The method is applicable to the arbitrary potentials of the form $V_q(r) = \sum_{n=0}^{\infty} g_n/(\alpha + \operatorname{ch} r)^n$.

1. INTRODUCTION

The perturbed Sine–Gordon equation

$$\partial_x^2 \varphi - \partial_t^2 \varphi = \phi_k(\varphi), \quad \phi_k(\varphi) = \sin \varphi + k \sin \frac{\varphi}{2}, \quad k \leq 1 \quad (1.1)$$

introduces, in a natural way, the binding of the Sine–Gordon soliton-soliton pairs [1]. Their perturbative description may be based on the exact static ground-state formula $\varphi = U_k = 4 \arcsin (\operatorname{ch} r + \alpha)^{-1/2}$, $\alpha > 0$, $r = \operatorname{const.} x$ [2]. In the lowest order, we put $\varphi = U_k + \Delta\varphi$, $\Delta\varphi = \sum_n \psi_n \exp(i\omega_n t)$ and get from (1.1) the linearized Schrödinger-like equation

$$-\frac{d^2}{dr^2} \psi(r) + V_q(r)\psi(r) = E\psi(r), \quad q = 2, \quad V_2(r) = -\frac{g}{\alpha + \operatorname{ch} r} - \frac{h}{(\alpha + \operatorname{ch} r)^2} \quad (1.2)$$

for the frequencies $\omega_n^2 = E + \operatorname{const.}$ and the Fourier components $\psi_n = \psi(r)$ of the excited-state correction $\Delta\varphi$. It is worth mentioning also that the other deformations ($\phi_k(\varphi) = \operatorname{sn}(\varphi/2, k) \operatorname{cn}(\varphi/2, k)/\operatorname{dn}^3(\varphi/2, k)$ [2]) of the Sine–Gordon interaction term ϕ_0 lead to the same form (1.2). This equation itself finds further application in the discussion of the stability or within the quantization formalism for solitons [3]. Even in the old quantum-mechanical Schrödinger equation interpretation, the potential well of the type $V_2(r)$ with the ‘core’ or two ‘barriers’ may be of interest in the molecular physics, etc.

We shall present here the compact and exact power-series solution of (1.2) which may easily be extended to any order q of the polynomial $V_q(r) = f(1/(\alpha + \operatorname{ch} r))$. In brief, we arrange both the powers of the independent variable and the coefficients into the q -dimensional vectors and define the generalized power series by means of their scalar products. In detail, we shall consider the bound-state solutions only.

2. THE HYPERGEOMETRIC EXAMPLE ($q = 1$)

From the formal point of view, Equation (1.2) with the special $\alpha = 1, h = 0$ case of $V_2(r)$ (the ‘modified Pöschl-Teller’ [4] or ‘Sine–Gordon-shaped’ [2] potential) coincides with the Gauss hypergeometric equation. On this example with $V_q(r) = -\lambda(\lambda - 1)/2(1 + \text{ch } r)$, $E = \kappa^2$, $q = 1$ and the $p = 0$ or $p = 1$ fundamental solutions

$$\psi(r) = \text{const. th}^p \frac{r}{2} \text{ch}^{-2\kappa} \frac{r}{2} {}_2F_1\left(\kappa + \frac{p}{2} + \frac{\lambda}{2}, \kappa + \frac{p}{2} + \frac{1-\lambda}{2}, 1 + 2\kappa, \frac{1}{\text{ch}^2(r/2)}\right) \quad (2.1)$$

we shall illustrate the overall structure of the method and compare the three methodical alternatives of Morse and Feshbach [5] (A), Flugge [4] (B), and the present one (C), respectively.

(A) The variable transformation $r \rightarrow z = (1 + \text{th}(r/2))/2$ in (1.2) implies the convergence of the hypergeometric series solution ${}_2F_1(\dots, z)$ in the full region $r \in (-\infty, \infty)$. When the asymptotically wrong $e^{+\kappa r}$ components are eliminated, the definite-parity requirement defines the energy levels $E = -\kappa_n^2$, $\kappa_n = \frac{1}{2}(\lambda - 1 - n)$, $n = 0, 1, \dots, n_{\max}$, $n_{\max} = \text{integer part of } (\lambda - 1)$. By chance, the resulting bound-state solution

$$\psi_n(r) = \text{const. ch}^{n+1-\lambda} \frac{r}{2} {}_2F_1\left(2\lambda - n - 1, -n, \lambda - n, \frac{1 + \text{th}(r/2)}{2}\right) \quad (2.2)$$

is polynomial – this is characteristic for the so called Bargman potentials [6].

(B) The initial choice of the variable $z = -\text{sh}^2(r/2)$ leads to the correct parity of solution from the very beginning. We must extend its convergence to the whole real axis $r \in (-\infty, \infty)$ by analytic continuation. Finally, the asymptotic boundary conditions quantize the energy levels.

(C) In our approach, we satisfy first the asymptotic conditions at $r \rightarrow \pm\infty$ by $z = 1/\text{ch}^2(r/2) = 2(\alpha + \text{ch } r)$, $\alpha = 1$. The series (2.1) is convergent in the half-interval $r \in (0, \infty)$ so that it is sufficient to match the two $r > 0$ and $r < 0$ branches together at $r = 0$.

Remark 1. The values $p = 0$ or $p = 1$ correspond to the solutions of the even or odd parity, respectively.

Proof by analytic continuation is easy. □

Remark 2. The matching (quantization) conditions read

$$\lim_{r \rightarrow 0} \psi'(r) = 0 \quad \text{or} \quad \lim_{r \rightarrow 0} \psi(r) = 0 \quad (2.3)$$

for $p = 0$ or 1 , respectively, and are numerically well behaved.

Proof. Let us fix some small $r \neq 0$ and replace (2.3) by the equivalent relations

$${}_2F_1\left(\kappa + \frac{\lambda}{2} + 1, \kappa + \frac{1-\lambda}{2} + 1, 2 + 2\kappa, \frac{1}{\text{ch}^2(r/2)}\right) \left(\kappa + \frac{1-\lambda}{2}\right) = 0, \quad (2.4)$$

or

$${}_2F_1\left(\kappa + \frac{\lambda}{2} + \frac{1}{2}, \kappa + \frac{1-\lambda}{2} + \frac{1}{2}, 1 + 2\kappa, \frac{1}{\text{ch}^2(r/2)}\right) = 0,$$

respectively. For $r \neq 0$ the left-hand side sums are convergent and may be approximated with an arbitrary precision by the N th partial sums, $N > N_0(r) \gg 1$. We put $\epsilon = \kappa - \kappa_n$, $|\epsilon| < \frac{1}{4}$, $n = 2M - p$, and decompose ${}_2F_1(\dots) \doteq \Sigma_0^{M-1} + \Sigma_M^N = f_1 + \epsilon f_2$ in (2.4). The functions $f_i = f_i(\epsilon)$, $i = 1, 2$ are varying slowly with ϵ . We define the approximate root $\epsilon = \epsilon(r, N)$ of (2.4) either by the transcendental equation $\epsilon = -f_1(\epsilon)/f_2(\epsilon)$ or by its simplified approximation $\epsilon = -f_1(0)/f_2(0)$. Finally, we find from the criterion of Raabe that the infinite series ${}_2F_1(\dots)$ in (2.4) is divergent at $r = 0$ so that $\epsilon(0, N) = 0(1/N)$ and $\lim_{r \rightarrow 0} \lim_{N \rightarrow \infty} \epsilon(r, N) = 0$. $\square$

3. THE EXACT SOLUTION ($q = 2$)

With an integer variable p equal to zero or one, let us introduce the two 2×2 matrices A_k and B_k .

$$\begin{aligned} A_k^{11} &= (2k-3)(2\kappa+2k-3) \\ A_k^{12} &= g - \alpha(\kappa+p+2k-3)(2\kappa+4k-5) \\ A_k^{21} &= 0 \\ A_k^{22} &= (2k-2)(2\kappa+2k-2) \\ B_k^{11} &= h + (\alpha^2 - 1)(\kappa+p+2k-3)(\kappa+p+2k-2) \\ B_k^{12} &= 0 \\ B_k^{21} &= g - \alpha(\kappa+p+2k-2)(2\kappa+4k-3) \\ B_k^{22} &= h + (\alpha^2 - 1)(\kappa+p+2k-2)(\kappa+p+2k-1), \quad k = 1, 2, \dots, \end{aligned} \quad (3.1)$$

and the two sets of functions.

$$\langle r | X_k^i \rangle = \text{sh}^p r / (\alpha + \text{ch } r)^{-4+\kappa+2k+i+p}, \quad (3.2)$$

where $k = 1, 2, \dots$, $i = 1$ or 2 , and

$$\langle r | Y_N^p \rangle = \sum_{k=1}^N \sum_{i=1}^2 \langle r | X_k^i \rangle D_k^i, \quad N \leq \infty \quad (3.3)$$

$$D_1^0 = 0, D_1^1 = 1, \quad D_k^i = - \sum_{j,m=1}^2 D_{k-1}^j B_{k-1}^{im} (A_k^{-1})^{mi}, \quad k = 2, 3, \dots$$

LEMMA 1. The action of the Hamiltonian $H = -(\text{d}^2/\text{d}r^2) + V_2(r)$ on the sets of functions $|X_k^i\rangle$ and $|Y_N\rangle = |Y_N^p\rangle$ satisfies the relations ($E = -\kappa^2$)

$$(E - H)|X_k^i\rangle = \sum_{j=1}^2 |X_k^j\rangle A_k^{ij} + \sum_{j=1}^2 |X_{k+1}^j\rangle B_k^{ij} \quad (3.4)$$

and

$$(E - H) |Y_N\rangle = \sum_{i,j=1}^2 |X_1^i\rangle A_1^{ij} D_1^i + \sum_{i,j=1}^2 |X_{N+1}^i\rangle B_N^{ij} D_N^i \quad (3.5)$$

respectively.

Proof. By insertion. □

LEMMA 2. *In the limit $N \rightarrow \infty$, the right-hand side of (3.5) vanishes for any $r \neq 0$.*

Proof. As a consequence of the definition (3.1), the first sum of the right-hand side of (3.5) is identically zero. Omitting the upper indices in the second one, we may write

$$D_{N_0+m} B_{N_0+m} \langle r | X_{N_0+m+1} \rangle = D_{N_0} B_{N_0} G^m \langle r | X_{N_0+1} \rangle z^{2m} \left(1 + 0 \left(\frac{1}{N_0} \right) \right), \quad (3.6)$$

$$G = -A_{N_0+1}^{-1} B_{N_0+1} \left(1 + 0 \left(\frac{1}{N_0} \right) \right) = U \begin{pmatrix} (1+\alpha)^2 & 0 \\ 0 & (1-\alpha)^2 \end{pmatrix} U^{-1}, \quad z = 1/(\alpha + \text{ch } r)$$

for any $m \geq 1$ and fixed $N_0 \gg 1$, some matrix $U \neq U(m)$ and zero limiting value of $(Gz^2)^m$ at $r \neq 0$ and $m \rightarrow \infty$. □

We may conclude that for $N \rightarrow \infty$ and $r \neq 0$, Equation (3.5) coincides with the Schrödinger Equation (1.2). The formal solution $|Y_N\rangle$ also exists in the limit $N \rightarrow \infty$:

LEMMA 3. *The infinite series $\langle r | Y_\infty \rangle$ is convergent for any $r \neq 0$.*

Proof. We have the 2×2 -matrix series (3.3)

$$\begin{aligned} \langle r | Y_\infty \rangle &= \dots + D_N \langle r | X_N \rangle + D_{N+1} \langle r | X_{N+1} \rangle + \dots \\ &= \dots + D_N (1 + Cz^2 + C^2 z^4 + \dots) \langle r | X_N \rangle, \end{aligned} \quad (3.7)$$

$$C = -B_{N-1} A_N^{-1} \left(1 + 0 \left(\frac{1}{N} \right) \right) = A_N G A_N^{-1} \left(1 + 0 \left(\frac{1}{N} \right) \right)$$

which may be diagonalized for $N \gg 1$ (cf., (3.6)). □

LEMMA 4. *In the point $r = 0$, the right-hand side of (3.5) remains bounded. The infinite-series part of the solution $\langle r | Y_\infty \rangle$ (i.e., $\lim_{r \rightarrow 0} \langle r | Y_\infty \rangle / \text{sh}^p r$) is convergent just for $p = 0$.*

Proof. Taking into account the next-order $O(1/N)$ corrections in the matrices $-A_N^{-1} B_N$ and $-B_{N-1} A_N^{-1}$ we are able to extend simply the proof of Lemmas 2 and 3 to $r = 0$ because the improved eigenvalues are

$$\left(\frac{1 \pm \alpha}{\text{ch } r + \alpha} \right)^2 \left(1 - \frac{\frac{3}{2} - P}{N} + 0 \left(\frac{1}{N^2} \right) \right). \quad \square \quad (3.8)$$

We see that the difference between (3.5) and (1.2) in the point $r = 0$ is inessential. Hence, we have

THEOREM. *The exact bound-state solutions of the Schrödinger Equation (1.2) are represented by the convergent power series $\psi(r) = \langle r | Y_\infty^p \rangle$ (3.3). Their parity is even or odd for $p = 0$ or $p = 1$, respectively, and the energies $E = -\kappa^2$ are determined by the matching conditions (2.3).*

Proof. It is sufficient to extend the proof of Remark 1 to $q = 2$. Considering the general properties [6] of (1.2) on the half-interval $r \in (0, \infty)$, we find that the independent solutions

$$\begin{aligned}\psi^{\text{even}}(r) &= C_1 \langle r | Y_\infty^0 \rangle + C_2 \langle r | Y_\infty^1 \rangle, \quad r > 0 \\ &= C_1 \langle r | Y_\infty^0 \rangle - C_2 \langle r | Y_\infty^1 \rangle, \quad r < 0 \\ \psi^{\text{odd}}(r) &= C_3 \langle r | Y_\infty^0 \rangle + C_4 \langle r | Y_\infty^1 \rangle, \quad r > 0 \\ &= -C_3 \langle r | Y_\infty^0 \rangle + C_4 \langle r | Y_\infty^1 \rangle, \quad r < 0\end{aligned}\tag{3.9}$$

are both regular at origin, i.e., all the necessary $r \rightarrow 0^\pm$ limits exist. Hence, we are permitted to require $\psi^{\text{even}}(0^+) = \psi^{\text{even}}(0^-)$ and $\psi^{\text{odd}}(0^\pm) = 0$, i.e.,

$$C_2 \langle r | Y_\infty^1 \rangle = 0, \quad C_4 \langle r | Y_\infty^1 \rangle = 0, \quad C_3 \langle r | Y_\infty^0 \rangle = 0, \quad r = 0.\tag{3.10}$$

Similarly, the continuity of the first derivative implies

$$C_1 \frac{d}{dr} \langle r | Y_\infty^0 \rangle = 0, \quad C_3 \frac{d}{dr} \langle r | Y_\infty^0 \rangle = 0, \quad C_2 \frac{d}{dr} \langle r | Y_\infty^1 \rangle = 0, \quad r = 0.\tag{3.11}$$

We see that any alternative to Remark 1, namely, the choice of $C_2 \neq 0$ or $C_3 \neq 0$, leads to the contradictory consequence $\langle 0 | Y_\infty^p \rangle = 0$ and $d/dr \langle 0 | Y_\infty^p \rangle = 0$, i.e., $\langle r | Y_\infty^p \rangle = 0$ for all r and $p = 1$ or 0 , respectively. $\square$

We note that the differentiation $d/dr \langle r | Y_\infty^p \rangle$ implies just the formal change $p \rightarrow p + 1$ in (3.8). Hence, the infinite series in (2.3) are all divergent at $r = 0$. In full analogy with Remark 2, we may, therefore, recommend again their explicitly differentiated, N -truncated forms with *fixed* $N \gg 1$ to be used in the effective numerical determination of $E = -\kappa^2$.

4. DISCUSSION

In the paper we have constructed the ‘minimal’ generalization of the hypergeometric function of one variable. Inspired by the practical needs, this function was found as an exact solution of the ordinary linear differential quasihypergeometric Equation (1.2) with ‘too many’ irreducible constants. The adequate method has proved to be the construction and explicit solution of the matrix two-term recurrences.

Besides the already mentioned Schrödinger and solitonic interpretations of (1.2), further possible physical applications of our solution may exist, especially in the fields where even the simplest models were rarely solvable. As an example, let us consider the static one-dimensional Klein–Gordon equation [7]

$$c^2 \hbar^2 \frac{d^2}{dr^2} \varphi(r) + [\epsilon - V_{q/2}(r)]^2 \varphi(r) = m^2 c^4 \varphi(r), \quad V_1(r) = -\frac{mG}{\alpha + \operatorname{ch} r}. \quad (4.1)$$

Parametrizing the bound-state energies $\epsilon = mc^2(1 - G\beta)c^2(1 + \alpha)$ by $\beta \in (0, 1)$ we transform (4.1) exactly into the Schrödinger-like form (1.2) with the assignment of parameters

$$\begin{aligned} -E = \kappa^2 &= \frac{m^2 G \beta}{\hbar^2 (1 + \alpha)} \left(2 - \frac{G \beta}{c^2 (1 + \alpha)} \right) > 0 \\ g &= \frac{2m^2 G}{\hbar^2} \left(1 - \frac{G \beta}{c^2 (1 + \alpha)} \right), \quad h = \frac{m^2 G}{c^2 \hbar^2}. \end{aligned} \quad (4.2)$$

The Klein–Gordon interpretation (4.1) of (1.2) also represents the strong motivation for introducing the generalizations

$$V_q(r) = -\sum_{n=1}^q \frac{g_n}{(\alpha + \operatorname{ch} r)^n} \quad (4.3)$$

of the potential $V_2(r)$. We would like to emphasize that the corresponding generalization of the $q = 2$ solution is entirely straightforward and the class of symmetric short-range potentials $V_q(r)$ becomes universal. Its semigroup property $V_q \times V_{q'} = V_{q+q'}$ enables us to define the simultaneous quasihypergeometric solutions of the one-dimensional nonrelativistic (Schrödinger) and relativistic (Klein–Gordon) equations with the same potentials (cf., [7]).

The three-dimensional s -wave Schrödinger equation differs from (1.2) just by the boundary condition $\langle r | Y_\infty \rangle = 0$ at $r = 0$. The corresponding solutions are equivalent to the present odd-parity ($p = 1$) subset of $\langle r | Y_\infty \rangle$'s. At the same time, the potential (4.3) may be related to the generalized Hulthén potentials [7], e.g., assuming $1 + \alpha \in (0, 1/\sqrt{2})$. Hence, the generalized hypergeometric solutions of Reference [7] may be identified with the analytic continuation of the class of quasihypergeometric functions $\langle r | Y_\infty^1 \rangle$. We note that in [7], the number M_k of the $\langle r | X_k^i \rangle$ - analogues of (3.2), $i = 1, 2, \dots, M_k$ increases with k . Here, it is fixed ($M_k = q$).

We may conclude that the generalized power-series method is able to inspire fruitful generalizations of the special functions: Besides the present quasihypergeometric case, its confluent forms are also introduced elsewhere [7, 8]. Altogether, it provides one of the desirable missing links between the two methodical extremes for the linear differential equations, namely, between their analytic and purely numerical treatments.

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